

- 6 (a) (i) to produce coherent sources or constant phase difference B1 [1]
- (ii) 1. $360^\circ / 2\pi$ rad allow $n \times 360^\circ$ or $n \times 2\pi$ (unit missing –1) B1 [1]
 2. $180^\circ / \pi$ rad allow $(n \times 360^\circ) - 180^\circ$ or $(n \times 2\pi) - \pi$ B1 [1]
- (iii) 1. waves overlap / meet B1
 (resultant) displacement is sum of displacements of each wave B1 [2]
 2. at P crest on trough (OWTTE) B1 [1]
- (b) $\lambda = ax / D$ C1
 $= 2 \times 2.3 \times 10^{-3} \times 0.25 \times 10^{-3} / 1.8$ C1
 $= 639 \text{ nm}$ A1 [3]